A LOCAL RING WITH A REGULAR SEQUENCE OF LENGTH THREE WHOSE QUOTIENT HAS A NONZERO ELEMENT KILLED BY TWO ENTRIES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: Hironaka-style example; the companion lemma lemma-base-change-regular-sequence consumes this construction.

Editorial note. The label lemma-strange-regular-sequence names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Section construction of the module E: E is the module of equivalence classes of pairs (f, g) with f in k[x, y, z, x inverse, y inverse] and g in k[x, y, z, y inverse], under the equivalence generated by f equals f-prime plus x h modulo y times k[x, y, z, x inverse] and g equals g-prime minus z h modulo y z times k[x, y, z]; the section prose proves (a) multiplication by x is injective on E, (b) multiplication by y is injective on E modulo xE, (c) E over the ideal generated by x and y is zero, and (d) the class of the pair (1, 0) yields a nonzero element delta in E over zE with x delta = y delta = 0
- Square-zero extension: for a ring S and an ideal I with I squared zero, the ring S plus I has maximal ideal the sum of the maximal ideal of S and I whenever that sum is an ideal
- Localization is exact, so a regular sequence remains regular after localizing at a multiplicative set

Lemma 0.1. *There exists a local ring R and a regular sequence x, y, z (in the maximal ideal) such that there exists a nonzero element $\delta \in R/zR$ with $x\delta = y\delta = 0$.*

Proof. Let $R = k[x, y, z] \oplus E$ where E is the module above considered as a square zero ideal. Then it is clear that x, y, z is a regular sequence in R, and that the element $\delta \in E/zE \subset R/zR$ gives an element with the desired properties. To get a local example we may localize R at the maximal ideal $\mathfrak{m} = (x, y, z, E)$. The sequence x, y, z remains a regular sequence (as localization is exact), and the element δ remains nonzero as it is supported at $\mathfrak{m}$. □

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